

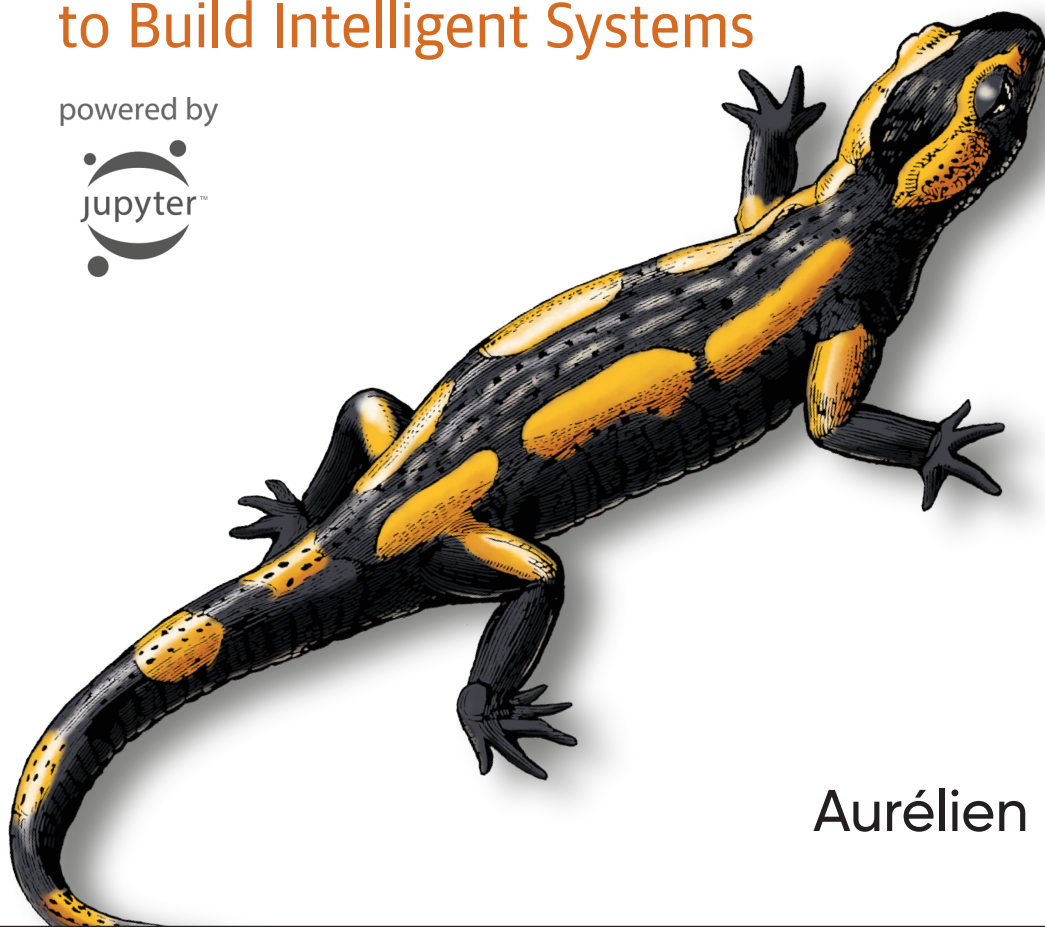
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Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques
to Build Intelligent Systems

powered by



Aurélien Géron

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

Through a series of breakthroughs, Deep Learning has boosted the entire field of machine learning. Now, even programmers who know close to nothing about this technology can use simple, efficient tools to implement programs capable of learning from data. The updated edition of this best-selling book uses concrete examples, minimal theory, and production-ready Python frameworks to help you gain an intuitive understanding of the concepts and tools for building intelligent systems.

You'll learn a range of techniques that you can quickly put to use. With exercises in each chapter to help you apply what you've learned, all you need is programming experience to get started. All code is available on GitHub. It has been updated to TensorFlow 2 and the latest version of Scikit-Learn.

- Learn Machine Learning fundamentals through an end-to-end project using Scikit-Learn and pandas
- Build and train many neural network architectures for classification and regression using TensorFlow 2
- Discover object detection, semantic segmentation, attention mechanisms, language models, GANs, and more
- Explore the Keras API, the official high-level API for TensorFlow 2
- Productionize TensorFlow models using TensorFlow's Data API, distribution strategies API, TF Transform, and TF-Serving
- Deploy on Google Cloud AI Platform or on mobile devices
- Exploit unsupervised learning techniques such as dimensionality reduction, clustering, and anomaly detection
- Create autonomous learning agents with Reinforcement Learning, including using the TF-Agents library

"An exceptional resource to study Machine Learning. You will find clear-minded, intuitive explanations, and a wealth of practical tips."

—François Chollet
Author of Keras, author of
Deep Learning with Python

"This book is a great introduction to the theory and practice of solving problems with neural networks; I recommend it to anyone interested in learning about practical ML."

—Pete Warden
Mobile Lead for TensorFlow

Aurélien Géron is a machine learning consultant and trainer. A former Googler, he led YouTube's video classification team from 2013 to 2016. He was also a founder and CTO of Wifirst (a leading Wireless ISP in France) from 2002 to 2012.

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SECOND EDITION

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

*Concepts, Tools, and Techniques to
Build Intelligent Systems*

Aurélien Géron

Beijing • Boston • Farnham • Sebastopol • Tokyo

O'REILLY®

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

by Aurélien Geron

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The Machine Learning Tsunami

In 2006, Geoffrey Hinton et al. published [a paper](#)¹ showing how to train a deep neural network capable of recognizing handwritten digits with state-of-the-art precision (>98%). They branded this technique “Deep Learning.” A deep neural network is a (very) simplified model of our cerebral cortex, composed of a stack of layers of artificial neurons. Training a deep neural net was widely considered impossible at the time,² and most researchers had abandoned the idea in the late 1990s. This paper revived the interest of the scientific community, and before long many new papers demonstrated that Deep Learning was not only possible, but capable of mind-blowing achievements that no other Machine Learning (ML) technique could hope to match (with the help of tremendous computing power and great amounts of data). This enthusiasm soon extended to many other areas of Machine Learning.

A decade or so later, Machine Learning has conquered the industry: it is at the heart of much of the magic in today’s high-tech products, ranking your web search results, powering your smartphone’s speech recognition, recommending videos, and beating the world champion at the game of Go. Before you know it, it will be driving your car.

Machine Learning in Your Projects

So, naturally you are excited about Machine Learning and would love to join the party!

1 Geoffrey E. Hinton et al., “A Fast Learning Algorithm for Deep Belief Nets,” *Neural Computation* 18 (2006): 1527–1554.

2 Despite the fact that Yann LeCun’s deep convolutional neural networks had worked well for image recognition since the 1990s, although they were not as general-purpose.

Perhaps you would like to give your homemade robot a brain of its own? Make it recognize faces? Or learn to walk around?

Or maybe your company has tons of data (user logs, financial data, production data, machine sensor data, hotline stats, HR reports, etc.), and more than likely you could unearth some hidden gems if you just knew where to look. With Machine Learning, you could accomplish the following **and more**:

- Segment customers and find the best marketing strategy for each group.
- Recommend products for each client based on what similar clients bought.
- Detect which transactions are likely to be fraudulent.
- Forecast next year's revenue.

Whatever the reason, you have decided to learn Machine Learning and implement it in your projects. Great idea!

Objective and Approach

This book assumes that you know close to nothing about Machine Learning. Its goal is to give you the concepts, tools, and intuition you need to implement programs capable of *learning from data*.

We will cover a large number of techniques, from the simplest and most commonly used (such as Linear Regression) to some of the Deep Learning techniques that regularly win competitions.

Rather than implementing our own toy versions of each algorithm, we will be using production-ready Python frameworks:

- **Scikit-Learn** is very easy to use, yet it implements many Machine Learning algorithms efficiently, so it makes for a great entry point to learning Machine Learning. It was created by David Cournapeau in 2007, and is now led by a team of researchers at the French Institute for Research in Computer Science and Automation (Inria).
- **TensorFlow** is a more complex library for distributed numerical computation. It makes it possible to train and run very large neural networks efficiently by distributing the computations across potentially hundreds of multi-GPU (graphics processing unit) servers. TensorFlow (TF) was created at Google and supports many of its large-scale Machine Learning applications. It was open sourced in November 2015, and version 2.0 was released in September 2019.
- **Keras** is a high-level Deep Learning API that makes it very simple to train and run neural networks. It can run on top of either TensorFlow, Theano, or Microsoft Cognitive Toolkit (formerly known as CNTK). TensorFlow comes with its

own implementation of this API, called *tf.keras*, which provides support for some advanced TensorFlow features (e.g., the ability to efficiently load data).

The book favors a hands-on approach, growing an intuitive understanding of Machine Learning through concrete working examples and just a little bit of theory. While you can read this book without picking up your laptop, I highly recommend you experiment with the code examples available online as Jupyter notebooks at <https://github.com/ageron/handson-ml2>.

Prerequisites

This book assumes that you have some Python programming experience and that you are familiar with Python's main scientific libraries—in particular, **NumPy**, **pandas**, and **Matplotlib**.

Also, if you care about what's under the hood, you should have a reasonable understanding of college-level math as well (calculus, linear algebra, probabilities, and statistics).

If you don't know Python yet, <http://learnpython.org/> is a great place to start. The official tutorial on **Python.org** is also quite good.

If you have never used Jupyter, **Chapter 2** will guide you through installation and the basics: it is a powerful tool to have in your toolbox.

If you are not familiar with Python's scientific libraries, the provided Jupyter notebooks include a few tutorials. There is also a quick math tutorial for linear algebra.

Roadmap

This book is organized in two parts. **Part I, *The Fundamentals of Machine Learning***, covers the following topics:

- What Machine Learning is, what problems it tries to solve, and the main categories and fundamental concepts of its systems
- The steps in a typical Machine Learning project
- Learning by fitting a model to data
- Optimizing a cost function
- Handling, cleaning, and preparing data
- Selecting and engineering features
- Selecting a model and tuning hyperparameters using cross-validation
- The challenges of Machine Learning, in particular underfitting and overfitting (the bias/variance trade-off)

- The most common learning algorithms: Linear and Polynomial Regression, Logistic Regression, k-Nearest Neighbors, Support Vector Machines, Decision Trees, Random Forests, and Ensemble methods
- Reducing the dimensionality of the training data to fight the “curse of dimensionality”
- Other unsupervised learning techniques, including clustering, density estimation, and anomaly detection

Part II, Neural Networks and Deep Learning, covers the following topics:

- What neural nets are and what they’re good for
- Building and training neural nets using TensorFlow and Keras
- The most important neural net architectures: feedforward neural nets for tabular data, convolutional nets for computer vision, recurrent nets and long short-term memory (LSTM) nets for sequence processing, encoder/decoders and Transformers for natural language processing, autoencoders and generative adversarial networks (GANs) for generative learning
- Techniques for training deep neural nets
- How to build an agent (e.g., a bot in a game) that can learn good strategies through trial and error, using Reinforcement Learning
- Loading and preprocessing large amounts of data efficiently
- Training and deploying TensorFlow models at scale

The first part is based mostly on Scikit-Learn, while the second part uses TensorFlow and Keras.



Don’t jump into deep waters too hastily: while Deep Learning is no doubt one of the most exciting areas in Machine Learning, you should master the fundamentals first. Moreover, most problems can be solved quite well using simpler techniques such as Random Forests and Ensemble methods (discussed in [Part I](#)). Deep Learning is best suited for complex problems such as image recognition, speech recognition, or natural language processing, provided you have enough data, computing power, and patience.

Changes in the Second Edition

This second edition has six main objectives:

1. Cover additional ML topics: more unsupervised learning techniques (including clustering, anomaly detection, density estimation, and mixture models); more techniques for training deep nets (including self-normalized networks); additional computer vision techniques (including Xception, SNet, object detection with YOLO, and semantic segmentation using R-CNN); handling sequences using convolutional neural networks (CNNs, including WaveNet); natural language processing using recurrent neural networks (RNNs), CNNs, and Transformers; and GANs.
2. Cover additional libraries and APIs (Keras, the Data API, TF-Agents for Reinforcement Learning) and training and deploying TF models at scale using the Distribution Strategies API, TF-Serving, and Google Cloud AI Platform. Also briefly introduce TF Transform, TFLite, TF Addons/Seq2Seq, and TensorFlow.js.
3. Discuss some of the latest important results from Deep Learning research.
4. Migrate all TensorFlow chapters to TensorFlow 2, and use TensorFlow's implementation of the Keras API (tf.keras) whenever possible.
5. Update the code examples to use the latest versions of Scikit-Learn, NumPy, pandas, Matplotlib, and other libraries.
6. Clarify some sections and fix some errors, thanks to plenty of great feedback from readers.

Some chapters were added, others were rewritten, and a few were reordered. See <https://homl.info/changes2> for more details on what changed in the second edition.

Other Resources

Many excellent resources are available to learn about Machine Learning. For example, Andrew Ng's [ML course on Coursera](#) is amazing, although it requires a significant time investment (think months).

There are also many interesting websites about Machine Learning, including of course Scikit-Learn's exceptional [User Guide](#). You may also enjoy [Dataquest](#), which provides very nice interactive tutorials, and ML blogs such as those listed on [Quora](#). Finally, the [Deep Learning website](#) has a good list of resources to check out to learn more.

There are many other introductory books about Machine Learning. In particular:

- Joel Grus’s *Data Science from Scratch* (O’Reilly) presents the fundamentals of Machine Learning and implements some of the main algorithms in pure Python (from scratch, as the name suggests).
- Stephen Marsland’s *Machine Learning: An Algorithmic Perspective* (Chapman & Hall) is a great introduction to Machine Learning, covering a wide range of topics in depth with code examples in Python (also from scratch, but using NumPy).
- Sebastian Raschka’s *Python Machine Learning* (Packt Publishing) is also a great introduction to Machine Learning and leverages Python open source libraries (Pylearn 2 and Theano).
- François Chollet’s *Deep Learning with Python* (Manning) is a very practical book that covers a large range of topics in a clear and concise way, as you might expect from the author of the excellent Keras library. It favors code examples over mathematical theory.
- Andriy Burkov’s *The Hundred-Page Machine Learning Book* is very short and covers an impressive range of topics, introducing them in approachable terms without shying away from the math equations.
- Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin’s *Learning from Data* (AMLBook) is a rather theoretical approach to ML that provides deep insights, in particular on the bias/variance trade-off (see [Chapter 4](#)).
- Stuart Russell and Peter Norvig’s *Artificial Intelligence: A Modern Approach*, 3rd Edition (Pearson), is a great (and huge) book covering an incredible amount of topics, including Machine Learning. It helps put ML into perspective.

Finally, joining ML competition websites such as [Kaggle.com](#) will allow you to practice your skills on real-world problems, with help and insights from some of the best ML professionals out there.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.



This element signifies a tip or suggestion.



This element signifies a general note.



This element indicates a warning or caution.

Code Examples

There is a series of Jupyter notebooks full of supplemental material, such as code examples and exercises, available for download at <https://github.com/ageron/handson-ml2>.

Some of the code examples in the book leave out repetitive sections or details that are obvious or unrelated to Machine Learning. This keeps the focus on the important parts of the code and saves space to cover more topics. If you want the full code examples, they are all available in the Jupyter notebooks.

Note that when the code examples display some outputs, these code examples are shown with Python prompts (`>>>` and `...`), as in a Python shell, to clearly distinguish the code from the outputs. For example, this code defines the `square()` function, then it computes and displays the square of 3:

```
>>> def square(x):  
...     return x ** 2  
...  
>>> result = square(3)  
>>> result  
9
```

When code does not display anything, prompts are not used. However, the result may sometimes be shown as a comment, like this:

```
def square(x):  
    return x ** 2  
  
result = square(3) # result is 9
```

Using Code Examples

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PART I

The Fundamentals of Machine Learning

The Machine Learning Landscape

When most people hear “Machine Learning,” they picture a robot: a dependable butler or a deadly Terminator, depending on who you ask. But Machine Learning is not just a futuristic fantasy; it’s already here. In fact, it has been around for decades in some specialized applications, such as Optical Character Recognition (OCR). But the first ML application that really became mainstream, improving the lives of hundreds of millions of people, took over the world back in the 1990s: the *spam filter*. It’s not exactly a self-aware Skynet, but it does technically qualify as Machine Learning (it has actually learned so well that you seldom need to flag an email as spam anymore). It was followed by hundreds of ML applications that now quietly power hundreds of products and features that you use regularly, from better recommendations to voice search.

Where does Machine Learning start and where does it end? What exactly does it mean for a machine to *learn* something? If I download a copy of Wikipedia, has my computer really learned something? Is it suddenly smarter? In this chapter we will start by clarifying what Machine Learning is and why you may want to use it.

Then, before we set out to explore the Machine Learning continent, we will take a look at the map and learn about the main regions and the most notable landmarks: supervised versus unsupervised learning, online versus batch learning, instance-based versus model-based learning. Then we will look at the workflow of a typical ML project, discuss the main challenges you may face, and cover how to evaluate and fine-tune a Machine Learning system.

This chapter introduces a lot of fundamental concepts (and jargon) that every data scientist should know by heart. It will be a high-level overview (it’s the only chapter without much code), all rather simple, but you should make sure everything is crystal clear to you before continuing on to the rest of the book. So grab a coffee and let’s get started!



If you already know all the Machine Learning basics, you may want to skip directly to **Chapter 2**. If you are not sure, try to answer all the questions listed at the end of the chapter before moving on.

What Is Machine Learning?

Machine Learning is the science (and art) of programming computers so they can *learn from data*.

Here is a slightly more general definition:

[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

—Arthur Samuel, 1959

And a more engineering-oriented one:

A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

—Tom Mitchell, 1997

Your spam filter is a Machine Learning program that, given examples of spam emails (e.g., flagged by users) and examples of regular (nospam, also called “ham”) emails, can learn to flag spam. The examples that the system uses to learn are called the *training set*. Each training example is called a *training instance* (or *sample*). In this case, the task T is to flag spam for new emails, the experience E is the *training data*, and the performance measure P needs to be defined; for example, you can use the ratio of correctly classified emails. This particular performance measure is called *accuracy*, and it is often used in classification tasks.

If you just download a copy of Wikipedia, your computer has a lot more data, but it is not suddenly better at any task. Thus, downloading a copy of Wikipedia is not Machine Learning.

Why Use Machine Learning?

Consider how you would write a spam filter using traditional programming techniques (**Figure 1-1**):

1. First you would consider what spam typically looks like. You might notice that some words or phrases (such as “4U,” “credit card,” “free,” and “amazing”) tend to come up a lot in the subject line. Perhaps you would also notice a few other patterns in the sender’s name, the email’s body, and other parts of the email.

2. You would write a detection algorithm for each of the patterns that you noticed, and your program would flag emails as spam if a number of these patterns were detected.
3. You would test your program and repeat steps 1 and 2 until it was good enough to launch.

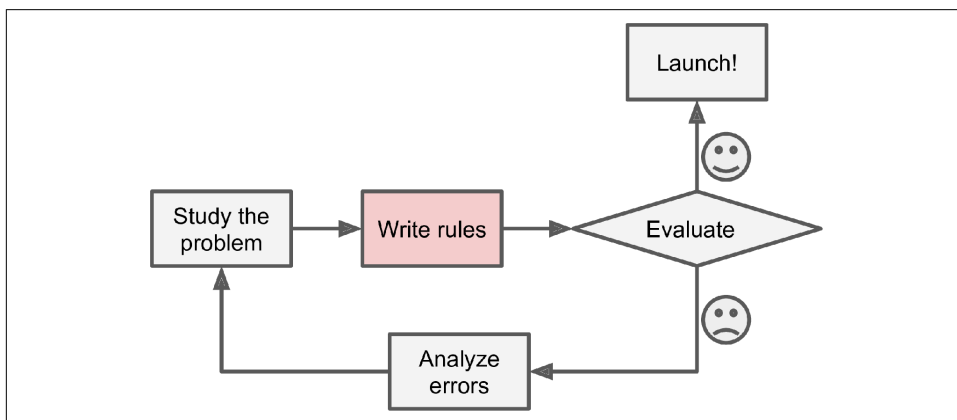


Figure 1-1. The traditional approach

Since the problem is difficult, your program will likely become a long list of complex rules—pretty hard to maintain.

In contrast, a spam filter based on Machine Learning techniques automatically learns which words and phrases are good predictors of spam by detecting unusually frequent patterns of words in the spam examples compared to the ham examples (Figure 1-2). The program is much shorter, easier to maintain, and most likely more accurate.

What if spammers notice that all their emails containing “4U” are blocked? They might start writing “For U” instead. A spam filter using traditional programming techniques would need to be updated to flag “For U” emails. If spammers keep working around your spam filter, you will need to keep writing new rules forever.

In contrast, a spam filter based on Machine Learning techniques automatically notices that “For U” has become unusually frequent in spam flagged by users, and it starts flagging them without your intervention (Figure 1-3).